SYNTHETIC DEMONSTRATION DATA. NOT VALID FOR REAL MACHINE OPERATION.

Standard Operating Procedure: Spindle & Coolant Controls

Page 1: Section 1: Scope, Approval & Revision Summary

Official Standard Operating Procedure for DEMO-CNC-01 Vertical Machining Center. Approved by Quality Engineering and Operations Director on 2024-01-15.

Revision B replaces and supersedes Rev A in its entirety. Spindle warmup cycle updated to dual-stage sequence to eliminate ceramic bearing thermal shock.

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Page 2: Section 2: Mandatory Spindle Warm-up Cycle (Approved)

Spindle Warm-up Procedure (Approved Rev B):
Stage 1: Run spindle at 1,500 RPM for exactly 5 minutes with flood coolant circulating.
Stage 2: Ramp spindle to 3,000 RPM for exactly 5 minutes.
Total warm-up duration: 10 minutes.

Coolant Concentration: Check coolant tank using optical refractometer every shift. Maintain 8.0% to 10.0% concentration for tough alloys (Titanium/Inconel). Minimum flood coolant pump pressure: 2.5 bar.

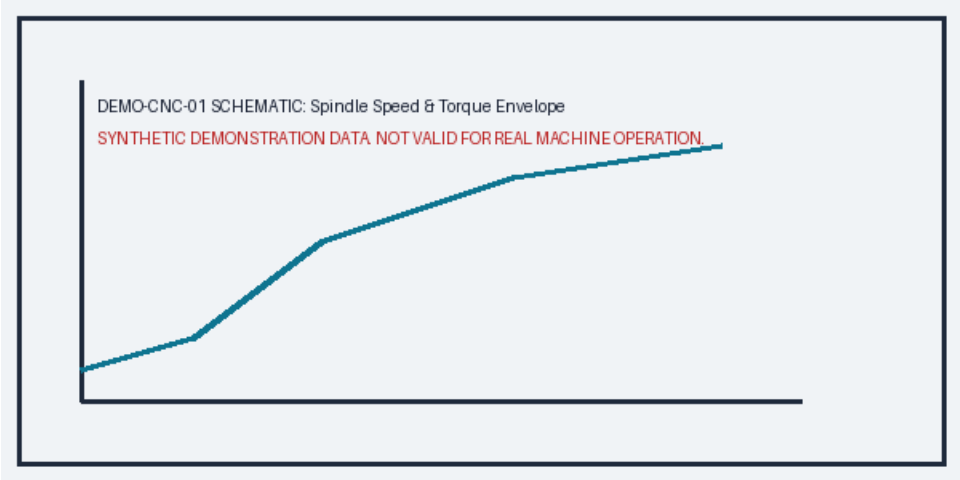


Figure 1: SOP-CNC-001 Operational Limits Diagram

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Page 3: Section 3: Catalog Feed & Speed References

Catalog Machining Parameters:
Aluminum 6061-T6: 10,000 RPM, feed rate 2,400 mm/min.
Titanium Grade 5 (Ti-6Al-4V) general milling catalog value: 3,500 RPM, feed rate 1,500 mm/min.
(Note: Always cross-reference active Quality Alerts before initiating cuts on exotic alloys).

Critical Safety Mandate: Enclosure doors must remain closed and interlocked during spindle rotation. Never bypass safety interlocks. Lockout/Tagout (LOTO) is mandatory before opening electrical cabinets.